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On the mechanism of sulfur fast diffusion in 3-D transition metals

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Abstract

The origin of abnormally fast diffusion of sulfur in face-centered cubic (fcc) nickel was investigated within an ab initio band structure calculation approach. A vacancy-mediated model of substitutional impurity diffusion with first-principles calculated parameters was used to determine the diffusion coefficients of S and Al impurities in fcc Ni. The sulfur diffusion coefficient was found to be two orders of magnitude higher than that of aluminum, in good agreement with experimental data. We demonstrate that ultrafast diffusivity of sulfur arises from its chemical interaction with the Ni matrix. The valence electron transfer from Ni3*d*–Ni3*d* bonds to sulfur softens the Ni–Ni bonding. That increases the sulfur–vacancy exchange rate, the sulfur–vacancy binding energy and the rotation rate of vacancy around sulfur. The latter is shown to be the key factor that governs S diffusion. We also discuss the impact of Ni–Ni bond softening on the mechanical properties of Ni.
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1. Introduction

Atomic diffusion controls the phase transformation and microstructure evolution of materials, especially at high temperatures. Despite the obvious fundamental and technological importance of diffusion in solids, we still know little about the microscopic mechanisms and factors that affect it. According to the traditional point of view, the strain induced by the size misfit between the host and the solute atoms plays a key role in the migration of substitutional impurities in metallic matrices [1]. A larger atom produces a larger lattice strain during elementary atom jump, which results in an increase in its migration barrier and ultimately lower diffusion mobility. This concept was challenged in Ref. [2], where is was shown by direct calculations that larger atoms in homologous series of 4*d* and 5*d* transition metals (TMs) have higher diffusion coefficients in face-centered cubic (fcc) Ni matrix. Another key factor responsible for the diffusion rate is the interaction between vacancies in the host metal and solute atoms [3], which is hypothesized to be controlled by electrostatic interactions [4–6] rather than by the size misfit. It has been shown [2] that, for TM solutes, the electronic structure of solute atoms – namely the *d*-state occupancy – plays a key role in the formation of migration barriers. This mechanism can clearly not be applied to the diffusion of *sp* substitutional elements (such as Al, S, P), and the role of the size misfit is unclear in this case.

Sulfur plays an important role in the mechanical properties of polycrystalline metals and metal–oxide interfaces [7,8]. Typically, the sulfur concentration in structural materials is very small (less than 10 ppma). However, sulfur diffuses up to three orders of magnitude faster than the other substitutional alloying elements in Ni-based alloys [9] (Fig. 1). Its diffusion to the internal interfaces and/or free

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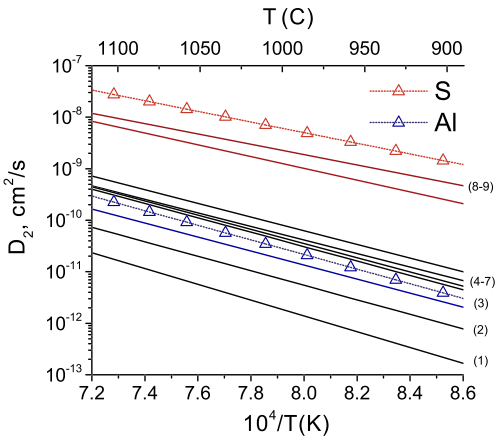


Fig. 1. The temperature dependence of diffusion coefficients for several solutes in Ni: 1 – Ni, 2 – Fe, 3 – Al, 4 – Sn, 5 – Sb, 6 – As, 7 – Te, 8 and 9 – S [10] and S [36]. Dotted lines with triangles represent calculation results of the present paper.

surfaces has a dramatic impact on the alloy's mechanical properties [10]. As an example, fast sulfur diffusion in Ni ($D_S/D_0 \sim 10^2$–10^3, where D_0 and D_S are the Ni self-diffusion and sulfur diffusion coefficients, respectively [11,12]) results in rapid kinetics of sulfur segregations formation [10,12], which promotes spallation of thermally grown oxide and thermal barrier coating failure [8]. Moreover, experimental data show [13] that fast sulfur diffusion takes place not only in the nickel matrix but is also typical for other TMs (see Fig. 2); the ratio D_S/D_0 varies from 10^2 to 10^3 for close-packed TM elements like Co, Ni and γ-Fe, and is about an order of magnitude larger than for Ag and Cu.

The reason for the fast sulfur diffusion in nickel is still controversial. Based on an experimental study of sulfur segregation due to quenched-in vacancies, Ferhat et al. [12] suggested that high sulfur diffusivity is mediated by solute–vacancy complexes, with the sulfur–vacancy binding energy being quite high, $E_{VB} = 1.5$–1.6 eV. Such a high value of E_{VB} seems to be unrealistic since typical impurity–vacancy binding energies do not exceed usually 0.4–0.6 eV [4,6]. Other experiments, on residual electric resistance measurements in binary Ni–S alloys after high-temperature electron irradiation [14], have suggested a sulfur diffusion mechanism in nickel that is mediated by

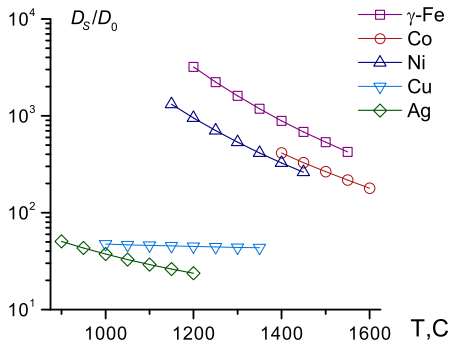


Fig. 2. Ratio of sulfur diffusion D_S and self-diffusion D_0 coefficients for selected transition metals vs. temperature [10,12].

vacancies rather than assisted by solute–vacancy complexes. Thus, atomic level modeling of the migration process is necessary to understand its physical origin.

First-principles density functional theory (DFT) provides an essential tool for uncovering microscopic mechanisms of the atomic migration in solids and predicting vacancy-mediated diffusion coefficients [2,15–22]. The diffusion of 4*d* and 5*d* TM impurities – from Zr to Ag, and from Hf to Au – in Ni matrix was investigated in Ref. [2] by combining the transition state model [23] with DFT calculations of migration barriers and vacancy–impurity binding energies. Simonovic and Sluiter concluded [22] that the electronic hybridization and magnetic polarization of impurity atoms play a major role in the diffusion process, rather than the atomic size mismatch or the vacancy–solute binding energy. The results of first-principles calculations of TM impurities diffusion in an Al host [16,22] did not reveal the inverse correlation between the atomic radius and the diffusion activation energy that was observed in Ref. [2]. It was concluded [16] that it happens because Al, unlike Ni, is a free-electron-like metal and the electrostatic interaction of impurity atom with positively charged vacancy makes the major contribution to the diffusion activation energies. Strong correlation between the diffusivities of TMs in Al and the occupation of *d*-states of TM impurities was observed [22], while no correlation with partial molar volume of TM impurities was found. Thus, recent theoretical results suggest that substitutional impurity diffusion process might be strongly affected by the change in electronic structure of both the host and the solute atoms.

In this paper we present the results of first-principles investigations of sulfur and aluminum diffusion in nickel. Al was chosen for comparative analysis as a common substitutional impurity in Ni (see Fig. 1), the diffusion coefficient of which is drastically different from that of sulfur. By using the vacancy-mediated diffusion model, we obtain diffusion coefficients of S and Al that are in good agreement with experimental results. We show that features of the electronic structure and chemical bonding in the vicinity of solute atoms play a crucial role in the enhanced diffusion of sulfur.

2. Methods of calculation

We employed a vacancy-mediated model of substitutional impurity diffusion [23,24] to calculate the diffusion coefficient of impurities in a solid solution (hereafter designated as D_2). Within this approach, D_2 can be expressed as

$$D_2 = D_0 \frac{f_2}{f_0}\frac{w_2}{w_0}\frac{w_4}{w_3} \approx D_0 \frac{f_2}{f_0}\frac{w_2}{w_0}\exp\left(\frac{E_{VB}}{kT}\right) \quad (1)$$

where D_0 is the host self-diffusion coefficient, E_{VB} is the impurity–vacancy binding energy, and f_0 and f_2 are the correlation factors for self-diffusion and impurity diffusion, correspondently. The value f_0 is equal to 0.78 for the fcc lattice and f_2 can be estimated through five atom jump frequencies, $w_{0,1,2,3,4}$ [24]:

$$f_2 = \frac{2w_1 + 7w_3F(w_4/w_0)}{2w_2 + 2w_1 + 7w_3F(w_4/w_0)}; \quad F(x) = 1 - \frac{10x^4 + 180.5x^3 + 927x^2 + 1341x}{7(2x^4 + 40.2x^3 + 254x^2 + 597x + 436)} \tag{2}$$

Here, w_0 (see Fig. 3) is the jump frequency of a vacancy in a pure matrix lattice (i.e. far from an impurity), w_1 is the jump frequency of a vacancy between one nearest-neighbor site of an impurity and another nearest-neighbor site, and w_2 is the same for an impurity–vacancy exchange; w_3 and w_4 are the dissociative jump frequency (when the vacancy goes away from the impurity) and reverse associative jump frequency, respectively. The frequencies of atomic jumps w_i can be written as

$$w_i = v_i \exp\left(\frac{\Delta S_i}{k}\right) \exp\left(-\frac{\Delta H_i}{kT}\right) \tag{3}$$

where v_i is an effective attempt frequency, which is related to atomic oscillation, and ΔS_i and ΔH_i are the entropy and enthalpy differences between the saddle point and the equilibrium atomic position for the ith type of atomic jump. We assume that the attempt frequencies of all host atoms are little affected by impurity and are equal, i.e. $v_0 \approx v_1 \approx v_3 \approx v_4$. The attempt frequency of the impurity may be estimated by [25]

$$\frac{v_2}{v_0} = \sqrt{\frac{m_0 T_{m2}}{m_2 T_{m0}}} \tag{4}$$

where v_0 is the value in the pure metal host, m_0 and m_2 are the masses of the host and impurity atoms, respectively, and T_{m0} and T_{m2} are the melting points. Following Ref. [25], we use the Wert–Zener relation to estimate the activation entropies:

$$\Delta S_i \approx \beta \frac{\Delta H_i}{T_{m0}} \tag{5}$$

Here, the factor β represents the temperature dependence of the shear modulus μ, $\beta = -\partial(\mu/\mu_0)/\partial(T/T_m)$, μ_0 being the shear modulus at absolute zero; $\beta \approx 0.12$ for the temperature range 1000–1300 K [26].Thus it is easy to show that the entropy contribution to the free energy is small, $T\Delta S_i \approx 10^{-1}\Delta H_i$. Moreover, the entropy contribution diminishes further when w_1/w_0 is calculated. Thus, only the activation enthalpies need to be calculated.

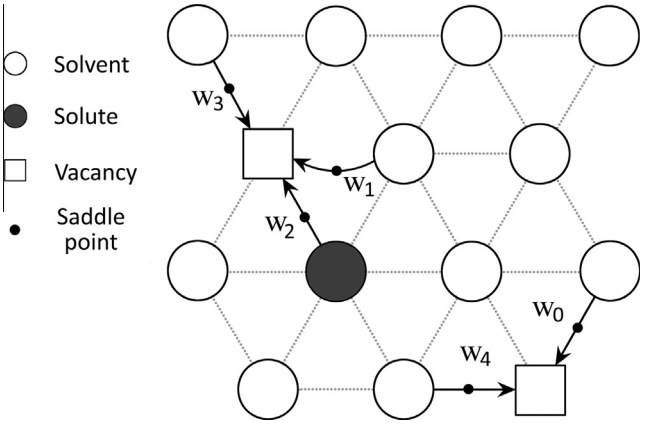


Fig. 3. Vacancy jumps near an impurity in fcc crystal.

In addition, we also calculate the effect of the impurity on the matrix self-diffusion coefficient. Due to the fact that some of the matrix atom jumps occur with altered frequency, the matrix self-diffusion coefficient could change. For small impurity concentrations ($C_S < 1\%$), this change is linear in concentration [27]:

$$D_0(C_S) = D_0(0)(1 + bC_S) \tag{6}$$

The quantity b is a self-diffusion enhancement factor, which can be expressed as

$$b = -18 + \frac{(4w_1\chi_1 + 14w_3\chi_2)}{w_0 f_0}\frac{w_4}{w_3} \approx -18 + \frac{(4w_1\chi_1 + 14w_3\chi_2)}{w_0 f_0}\exp\left(-\frac{E_{VB}}{kT}\right) \tag{7}$$

where χ_1 and χ_2 are partial correlation factors for different types of jumps and are functions of w_i (see Ref. [27] for more details).

To calculation the impurity–vacancy binding energies and migration enthalpies, we use the Vienna Ab-initio Simulation Package (VASP) package [28–30], with first-principles pseudopotentials constructed by the projected augmented wave (PAW) approach [31], and the generalized gradient approximation for the density functional in the form proposed by Perdew et al. [32] with the parameterization from Ref. [33]. The following calculation parameters are used to produce the convergence of migration barriers within 0.05 eV: PAW potentials; an energy cutoff of 400 eV; and the uniform k-point $4 \times 4 \times 4$ mesh in the Monkhorst-Park scheme [34]. We employed a trigonal 64-atom supercell derived from the fcc primitive cell $4 \times 4 \times 4$ expansion along all three translation vectors in an fcc lattice. The chosen supercell is large enough to eliminate any possible size effects, as this issue has been thoroughly addressed in the literature (see e.g. Refs. [2,18]).

The migration enthalpies, ΔH_i, represent the system enthalpy difference between the states in which the diffusing atom is located at its equilibrium lattice site and at the saddle point along the migration path. The saddle point positions were calculated by the nudged elastic band technique implemented in VASP.

In the present study, all of the calculations of ΔH_i were carried out at zero temperature with full lattice relaxation at a constant pressure (e.g. ionic, volume and cell shape relaxation). For the saddle point configurations, the position of diffusing atom was fixed, while the positions of other atoms were allowed to relax.

3. Results of calculations

Table 1 shows calculated values of ΔH_i for different types of jump. As can be seen, the migration enthalpy

Table 1
Calculated migration enthalpies ΔH_i (eV), correlation factor f_2, self-diffusion enhancement factor b, impurity diffusion coefficient $D_{2,\text{theor}}$ and experimental diffusion coefficient [11,13] $D_{2,\text{exp}}$ for $T = 1000$ °C.

	ΔH_0	ΔH_1	ΔH_2	ΔH_3	ΔH_4	f_2	b	$D_{2,\text{theor}}$ ($\text{cm}^2\,\text{s}^{-1}$)	$D_{2,\text{exp}}$ ($\text{cm}^2\,\text{s}^{-1}$)
Al	1.0	0.93	0.60	0.83	0.76	0.21	42	3.37×10^{-11}	2.1×10^{-11}
S	1.0	0.45	0.25	0.94	0.52	0.23	4.1×10^3	7.1×10^{-9}	$(1.5–2.6) \times 10^{-9}$

ΔH_i for Al atoms does not differ much from that of Ni, ΔH_0. Quite the opposite situation is observed in the case of S. Sulfur's migration enthalpy ΔH_2 is much lower than ΔH_0 and the barriers of the surrounding Ni atoms, with ΔH_1 being almost halved (see Table 1).

Solute–vacancy binding energies E_{VB} were calculated directly from first principles as the energy differences between two positions – next to the impurity and at the maximum possible distance from it within the computational cell (the seventh coordination sphere of impurity). We found energies of 0.42 and 0.09 eV for S and Al, respectively. In the nearest-neighbor (NN) approximation, the binding energies are equal to the difference $\Delta H_3 - \Delta H_4$, which is 0.41 eV for S and 0.1 eV for Al. This means that the NN approximation is sufficient for the systems under investigation.

The solute–vacancy binding energy for S is comparable to the results of the first-principles calculations for S in Fe [35], but contradicts the estimates of 1.5–1.6 eV presented in Ref. [12]. The reason for this discrepancy in as follows. In Ref. [12] the solute vacancy binding energy was calculated as a difference of the measured diffusion activation energy Q, the vacancy formation E_{VF} and vacancy migration E_{VM} energies: $E_{\text{VB}} = E_{\text{VF}} + E_{\text{VM}} - Q$. E_{VF} and E_{VM} were assumed to be 1.6 and 1.27 eV, respectively. Vacancy migration energy E_{VM} in such interpretation is equal to the ΔH_2 barrier. From Table 1 one can see that its real value is 1 eV lower than was assumed in Ref. [12], giving $E_{\text{VB}} \approx 0.5$ eV, which is more realistic.

The calculated diffusion coefficients for S and Al are presented in Table 1 for $T = 1000$ °C. The temperature dependence of the calculated diffusion coefficients is presented, together with experimental data, in Fig. 1 [13]. The Ni self-diffusion coefficient was approximated by the equation $D_0 = 2.59 \times \exp(-3.05\ \text{eV}/kT)\ \text{cm}^2\,\text{s}^{-1}$ [13]. In particular, we found the sulfur diffusion coefficient to be two orders of magnitude larger than that for aluminum. The calculated (dashed) curves in Fig. 1 can be approximated by the Arrhenius equation:

$$D_2 = D_{20} \exp\left(-\frac{Q}{kT}\right) \tag{8}$$

Table 2
Calculated and experimental (in parenthesis) apparent diffusion activation energies and pre-exponential factors for S and Al in Ni.

	D_{20} ($\text{cm}^2\,\text{s}^{-1}$)	Q (eV)
S	0.84 (1.4 [13], 0.2 [36])	2.04 (2.27 [13], 2.00 [36])
Al	4.3 (1.0 [13], 7.5 [37])	2.81 (2.7 [13], 2.95[37])

where Q is the apparent activation energy and D_{20} is a corresponding pre-exponential factor. We compare the calculated values of Q and D_{20} with experimental data in Table 2 and find a reasonably good agreement for both sulfur and aluminum impurities.

4. Discussion

The diffusivity calculation method used in the present study is widely accepted [2,16,18,22,25]. At the same time, in a more rigorous approach, the entropy contribution to the diffusion barrier can be treated within Vineyard's model [38], which gives the following expression for the pre-exponential factor in Eq. (3):

$$\nu_i^* = \nu_i \exp(\Delta S_i/k) = \left(\prod_{i=1}^{3N} \nu_i\right) \Big/ \left(\prod_{i=1}^{3N-1} \nu_i'\right) \tag{9}$$

The "effective" frequency ν_i^* within the classical harmonic approximation is the ratio of the product of the $3N$ normal frequencies of the system in the initial configuration to the $3N - 1$ normal frequencies of the system constrained in the saddle point configuration. However, the entropy term is usually less than $(1–2)k$ and constitutes only a minor effect, as was thoroughly investigated by Mantina et al. [18,19]. In addition, Rauls and Frauenheim [39] have shown that, in order to derive formation entropies of point defects, very large supercells on the order of a few hundred atoms are needed because the entropy converges slowly with supercell size. Supercells which are sufficiently large for this purpose are computationally too expensive for the present study. Therefore, we believe that estimation of the migrational entropy within the phenomenological approach of Eqs. (2) and (3) is quite reasonable and gives good agreement with direct experimental results. This shows that major physical factors for diffusion process are accounted for.

Let us now consider the microscopic mechanisms responsible for the differences in the S and Al diffusion processes and compare them to the $4d$ and $5d$ impurities in Ni studied by Janotti et al. [2]. For this, we rewrite Eqs. (1)–(3) in the following form:

$$D_2 = \frac{D_0}{f_0} \Gamma \exp\left\{\frac{E_{\text{VB}}}{k_B T}\right\}, \tag{10}$$

where

$$\Gamma = \frac{\Gamma_{\text{ex}}(\Gamma_{\text{rot}} + \Gamma_{\text{dis}})}{\Gamma_{\text{ex}} + (\Gamma_{\text{rot}} + \Gamma_{\text{dis}})}. \tag{11}$$

One can see that four parameters govern the diffusion rate, namely: the solute–vacancy binding energy E_{VB}: the solute–vacancy (S–V) exchange, $\Gamma_{ex} = w_2/w_0$; the S–V pair rotation, $\Gamma_{rot} = w_1/w_0$; and the dissociation rates, $\Gamma_{dis} = w_3/w_0$. All these rates are normalized by the Ni–vacancy exchange rate in pure Ni, w_0. The rate parameter Γ can be interpreted by analogy with the electrical conductivity of an equivalent electric circuit in which the parallel connection of Γ_{rot} and Γ_{dis} is connected in series with Γ_{ex}.

Following Janotti et al. [2], we consider two limiting cases. If S–V exchange is very slow, i.e. $\Gamma_{ex} \ll \Gamma_{rot} + \Gamma_{dis}$, Eq. (11) reduces to $\Gamma \approx \Gamma_{ex}$. This means that the diffusion is governed solely by solute jumps and its activation energy is equal to the sum of the solute migration energy, ΔH_2, and the solute–vacancy binding energy, E_{VB}, $Q \approx \Delta H_2 + E_{VB}$. This is exactly the case of Janotti's mid-row TM solutes (see the first row in Table 3).

In the case where the S–V exchange is very fast, i.e. $\Gamma_{ex} \gg \Gamma_{rot} + \Gamma_{dis}$, Eq. (11) reduces to $\Gamma \approx \Gamma_{rot} + \Gamma_{dis}$. The diffusion process is governed by the S–V pair rotation and dissociation rates. This is the case for Al diffusion (see Table 3). For the end-row elements, $\Gamma_{rot} \ll \Gamma_{dis}$; and the equation for Γ further reduces to $\Gamma \approx \Gamma_{dis}$. The diffusion activation energy in this case is equal to a sum of the S–V dissociation barrier ΔH_3 and the S–V binding energy E_{VB}, $Q \approx \Delta H_3 + E_{VB}$.

The key factor that controls the sulfur diffusion is different from all considered above cases. As demonstrated by row 4 of Table 1 for sulfur, both $\Gamma_{ex} \gg \Gamma_{dis} + \Gamma_{rot}$ and $\Gamma_{rot} \gg \Gamma_{dis}$. This means that sulfur diffusion is governed by the S–V pair rotation rate, $\Gamma \approx \Gamma_{rot}$, and $Q \approx \Delta H_1 + E_{VB}$. The activation energy for sulfur diffusion does not depend on the S migration barrier but is determined by the migration barrier of the neighboring Ni atoms. To estimate the error introduced by the assumption of equal attempt frequencies of host atoms in Eq. (3), $v_0 \approx v_1 \approx v_3 \approx v_4$, we calculate the change in the most affected by the Ni matrix softening ratio $v_1/v_0 \approx \sqrt{\Delta H_1/\Delta H_0}$, which contributes to S–V pair rotation rate. The corrected value of the S diffusion coefficient at 1000 °C is 2.6×10^{-9} cm^2 s^{-1} (compared to 3.6×10^{-9} cm^2 s^{-1} calculated under the assumption of equal attempt frequencies v_i). This qualitative estimate indicates that the impact of the change in the phonon spectrum on the S diffusion rate that is induced by the local Ni matrix softening in the vicinity of an S atom is insignificant compared to the major factors that can change the results by orders of magnitude.

Janotti et al. [2] suggested that diffusion is governed by the atomic size, i.e. large atoms diffuse faster. The size of the impurity in the metal host can be characterized by the size of its first coordination sphere, R. We performed the calculation of R for S, Al and a number of 4d TMs in Ni host using the ab initio technique described above. Table 4 shows that end-row elements are indeed up to 5% larger than mid-row elements and Ni itself, and they induce larger atomic distortions. Sulfur, on the other hand, is smaller than all 4d and 5d impurities and smaller than Al. Thus, the "large atoms diffuse faster" rule does not explain the faster sulfur diffusivity. This finding highlights the importance of the electronic structure features in understanding the diffusion of sulfur.

Fig. 4 shows the charge density distribution of valence electrons in Ni containing a vacancy and a solute atom in the Ni site (a and c) and in the saddle point position (b and d). The chemical bonding between solute and Ni atoms appears to be quite different for S and Al. We have calculated the atomic charges using the Bader method [40] and found them to be $Q_S = -0.4e$ for sulfur and $Q_{Al} = +1.9e$ for aluminum. Aluminum donates electrons to the Ni host, contributing an additional charge to the Ni–Ni covalent d-bonds and increasing the stiffness of the surrounding Ni matrix. This effect was discussed in Ref. [41] and appeared as a dramatic increase in the bulk modules of transition metals as the number of valence electrons increased. Grossman et al. [41] also found that in an Ni_3Al structure a depletion of electron density at the Al site is accompanied by a significant build-up of directional d-bonding charge between the nearest-neighbor Ni atoms.

Table 3
Contributions to the diffusion constant from different atomic processes.

Elements	Exchange w_2/w_0	Rotation w_1/w_0	Dissociation w_3/w_0	S–V bonding w_4/w_3	Γ	$\Gamma \cdot \exp(E_{VB}/kT)$
TMs, mid-row [2]	*from 0.2*	1	1	1	0.19	0.19
	to 0.003	1	1	1	0.003	0.003
TMs, end-row [2]	from 30	0.02	*1*	20	1.03	23.9
	to 3500	0.02	*1*	20	1.07	24.8
Al	16.3	*1.00*	*2.2*	1.8	3.8	6.9
S	178.6	*51.1*	1.2	15.2	40.6	616.7

The italicised cells indicate the governing processes.

Table 4
Calculated atomic distortion around the impurity a number of *sp*- and 4*d* impurities normalized.

Element	Al	S	Y	Zr	Tc	Ru	Ag	Cd
$(R_{imp} - R_{Ni})/R_{Ni} \times 100\%$	+0.35	+0.08	+5.1	+3.4	+0.41	+0.62	+3.1	+3.6

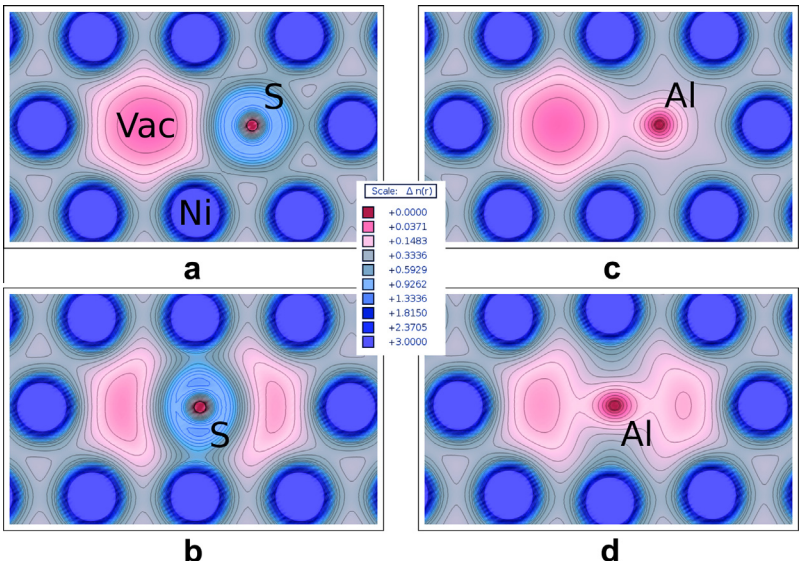


Fig. 4. Charge density contour map in the {1 1 1} Ni plane with S (a, b) and Al (c, d) solute atoms in the Ni site (a, c) and at the saddle point (b, d).

The effect of sulfur is quite the opposite. Sulfur removes valence *d*-electrons from neighboring Ni–Ni bonds to form ionic and covalent bonds between the S3*p* and Ni3*d* orbitals; more strong covalent bond formation takes place in the saddle point position (see Fig. 4a and b), where the coordination number decreases from 12 (in the initial position) to 4 (at the saddle point). The removal of *d*-electrons also leads to destabilization (or softening) of the surrounding Ni matrix (this effect was discussed for the first time in Ref. [42]). As shown in Ref. [43] by calculation of the adhesion properties of the Ni–oxide interface contaminated by S, decohesion occurs via the breaking of Ni–Ni and Ni–O bonds, with Ni–S bonds remaining unbroken.

The effect of sulfur on chemical bonding appears to be the main reason for its increased diffusivity. The enhancement of the S–Ni chemical bonding at the saddle point leads to a low sulfur migration barrier ΔH_2 due to the compensation of the lattice distortions around the sulfur atom (the first coordination sphere radii in the saddle point positions are 2.146 and 2.268 Å for S and Al, respectively). The softening of the Ni host around sulfur impurities results in a lower strain energy penalty for the adjacent Ni atom to move through the saddle point toward a vacancy, thus the energy barrier ΔH_1 is significantly reduced. We also conjecture that the higher sulfur–vacancy binding energy E_{VB} originates mainly from the Ni host softening and the electrostatic interaction between the electron excess of the sulfur atom and the electron deficiency in the vacancy region, since size misfit arguments are not applicable in this case.

It is reasonable to expect behavior similar to that of sulfur in other TMs. When S3*p*–Me3*d* hybridization is weak, the diffusivity of sulfur is expected to be much lower for metal matrices with completely filled or absent electron *d*-shells, such as Cu and Ag. Indeed, this phenomenon was observed experimentally (see Fig. 2). The same holds true for the case of diffusion of TMs in *sp* hosts [22]. In general, we suggest that the electronic effects play the major role in diffusion of *sp*/TM elements in TM/*sp* metal hosts, where there is a significant electron transfer from the host atoms to the impurity. At the same time, it seems reasonable to expect that diffusion of *sp*/TM elements in *sp*/TM metal hosts is primarily governed by atomic size misfit.

Reduction of the Ni migration barriers in the vicinity of S results in the enhancement of the self-diffusion coefficient of the alloy. This effect depends on the solute concentration and may play an important role in the process of mass transfer in the regions enriched by sulfur, such as grain boundaries, dislocations or crack tips. This may lead to an increased rate of crack initiation and propagation, and accelerated void growth.

Sulfur-enhanced Ni self-diffusion near the grain boundaries together with softening of the Ni lattice due to sulfur segregation may also be important factors in grain boundary brittleness and thermally grown oxide scale spallation [44]. This conclusion agrees with the previous results of calculations of sulfur stimulated decohesion for low sulfur concentrations at interfaces [43]. For high concentrations, the effects of sulfur aggregation and S–S interaction should also be important [7].

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